ABDULLAH OTHMAN

Electrical Engineering PhD | Digital Twins, Wireless Systems & Optimization

abdullah.b.othman@gmail.com +1 438 925 1017 Montréal, QC, Canada abdullah-b-othman Website

PROFILE

Ph.D. in Electrical Engineering focused on 5G-Advanced/6G wireless systems, radio resource management, digital twins, and non-terrestrial networks. Research combines mathematical optimization, AI/ML, ray-traced system-level modeling, and uncertainty-aware control for cellular, UAV/HAPS, and resilient IIoT networks. Developed digital-twin frameworks that couple learned outage detection and model synchronization with resource-allocation decisions, and NTN algorithms spanning trajectory design, beamforming, user association, positioning uncertainty, and finite backhaul. Strong MATLAB/Python implementation, simulation, technical writing, and IEEE publication experience. Holds four granted U.S. patents arising from deep-learning research. **Authorized to work in Canada.**

EXPERIENCE

Doctoral Researcher

École de technologie supérieure (ÉTS)

2021 – 2026 Montréal, QC

- Designed **radio-resource-management algorithms** for cellular/OFDMA and NB-IoT networks, including power control, subcarrier scheduling, user association, coverage enhancement, and interference management; also analyzed convergence behavior of iterative power-control methods.
- Developed **digital-twin-enabled wireless frameworks** integrating ray-traced channel modeling, model synchronization, and resource optimization for resilient and post-disaster network operation.
- Built a **learning-enabled outage-management pipeline** using SINR-labeled digital-twin data and EfficientNet-based inference to predict user outages in near real time; detector outputs trigger UAV-assisted uplink resource-allocation decisions.
- Developed **HAPS/UAV-assisted NTN optimization** methods coupling UAV trajectories, access/backhaul beamforming, user association, finite-backhaul constraints, and estimator-aware reliability under position uncertainty and heavy-tailed interference.
- Formulated and solved **nonconvex and mixed-integer wireless optimization** problems using SCA, difference-of-convex methods, SDR, Dinkelbach optimization, branch-and-bound, stochastic/risk-aware techniques, and Kalman-filter-based state estimation; implemented system-level simulations primarily in MATLAB and Python.

Electrical Engineer

Naizak Global Engineering Systems

2018 – 2021 Khobar, Saudi Arabia

- **Diagnosed, repaired, and validated multidisciplinary laboratory equipment** across KFUPM engineering departments, including control, flow and pressure, analytical, sensing, electronic, and laboratory-automation systems.
- Performed **circuit- and component-level fault isolation**, prepared inspection reports and corrective-action recommendations, and coordinated repairs, replacements, equipment relocation, and follow-up with laboratory users and vendors.

EDUCATION

Ph.D. Electrical Engineering

ÉTS

2021 – April 2026 Montréal

Thesis: Digital-Twin-Enabled and Risk-Aware Radio Resource Management for Resilient IIoT Smart Grid Communications.
Advisor: Prof. Georges Kaddoum.
Mention: Excellent (highest grade).

M.Sc. Telecommunications

KFUPM

2015 – 2018 Dhahran

B.Sc. Electrical Engineering

KFUPM

2011 – 2015 Dhahran

FOCUS AREAS

RRM | 5G-Advanced/6G | 3GPP NTN | NB-IoT/mMTC | UAV/HAPS | MIMO/Beamforming | Digital Twins | AI/ML for Wireless | Sensing/Positioning | System-Level Simulation

SKILLS

Wireless Modeling

OFDMA | Channel modeling | Ray tracing | State estimation (UKF)

Optimization

Convex optimization | Mixed-integer optimization | Robust/risk-aware opt. | Metaheuristics

Machine Learning

Deep learning | CNNs | Supervised learning

Programming

MATLAB | Python | C++ | LaTeX

Tools & Frameworks

CVXPY | MOSEK | Gurobi | Sionna RT | TensorFlow | scikit-learn | OpenCV

Research & Teaching Assistant

KFUPM

2015 – 2018 Dhahran, Saudi Arabia

- Conducted M.Sc. research on wireless-link design for seismic data acquisition, maximizing multi-access uplink sum rate using integer programming and metaheuristics; published at SEG 2018.
- Coauthored deep-learning research on passive seismic event detection and denoising published in IEEE TGRS and contributing to four granted U.S. patents.
- Taught circuits and electronics laboratories, led weekly experiments, prepared assignments, and graded coursework.

PUBLICATIONS

Summary: 6 journal papers, 2 conference papers, and 4 granted U.S. patents; one additional manuscript under review at IEEE TVT.

Selected Manuscript Under Review

- "Estimation-Coupled Risk-Aware Resource Allocation for HAPS–UAV Networks Under Impulsive Interference and Finite Backhaul." *Manuscript under review, IEEE Transactions on Vehicular Technology.*

Selected Journal Papers

- "Digital-Twin-Enabled Outage Detection and Uplink Availability Optimization With UAV Assistance". *IEEE Communications Letters*, 2026, 30: pp. 1255–1259.
- "Joint Digital Twin Synchronization Scheduling and Resource Allocation for Post-Disaster Wireless Networks". *IEEE Transactions on Network and Service Management*, 2026.
- "Joint Optimization of Radio Resources and Coverage Enhancement in Massive Microgrid Networks". *IEEE Transactions on Network and Service Management*, 2023, pp. 1–16.
- "Digital Twinning in Smart Grid Networks: Interplay, Resource Allocation and Use Cases". *IEEE Communications Magazine*, 2023, 61: pp. 120–126.
- "Automated Event Detection and Denoising Method for Passive Seismic Data Using Deep Neural Networks". *IEEE Transactions on Geoscience and Remote Sensing*, 2021, pp. 1–8.

Patents (all granted, U.S.)

- "Computer System for Event Detection of Passive Seismic Data". US 12,196,902 B2, Issued Jan. 14, 2025.
- "Apparatus and System for Identifying Seismic Features". US 12,117,577 B2, Issued Oct. 15, 2024.
- "Apparatus with Circuitry for Seismic Data Detection". US 12,105,236 B1, Issued Oct. 1, 2024.
- "Event Detection and De-noising Method for Passive Seismic Data". US 12,092,780 B2, Issued Sep. 17, 2024.

AWARDS

Mention Excellent — Ph.D. thesis
Highest grade conferred by ÉTS, May 2026.

Jury nomination — thesis excellence award
ÉTS, 2026.

University Scientific Forum
1st Place (Dept.) & 4th Overall, KFUPM, 2018

Letter of Commendation — Teaching Excellence
KFUPM Teaching Lab, 2017

1st Place, Short Novel Writing Competition
Jarir Reader platform, 2017.

FUNDING

Supported by Hydro-Québec & NSERC Alliance Grant (ALLRP 552856-20), 2021 – 2025. Previously supported by CeGP Energy and Geo-Processing Grant (GTEC1601, KFUPM & Georgia Tech), 2016 – 2018.

LANGUAGES

English (IELTS 8.0)
Arabic (Native)
French (B2)

SERVICE & VOLUNTEERING

Reviewer: IEEE OJCOM, IEEE TMC, IEEE TVT, IEEE TASE, IEEE TNSE, IEEE WCL, IEEE PIMRC, Springer J. Supercomputing.

October 2026 – present Private tutoring via Superprof: Calculus AB (differentiation) and electrical circuits; lesson preparation, tailored explanations and guided problem solving with a digital tablet.

October 2026 Research poster presentation, Hydro-Québec's Symposium 3i, IREQ, Varennes: digital twins and radio-resource allocation.

September 2026 Volunteer Translator, Collège de Rosemont. Translated French into Arabic to help new students understand college rules and procedures.

2024 Guest tutorial: "Digital Twins for Wireless Networks," Ph.D. seminar series, ÉTS, Apr. 2024. Covered digital-twin architectures, ray-tracing channel modeling, and smart-grid use cases.